

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A machine makes large boxes or small boxes, one at a time, for a total of **700** minutes each day. It takes the machine **10** minutes to make a large box or **5** minutes to make a small box. Which equation represents the possible number of large boxes, ***x***, and small boxes, ***y***, the machine can make each day?

Answer

- A. $5x + 10y = 700$
- B. $10x + 5y = 700$
- C. $(x + y)(10 + 5) = 700$
- D. $(10 + x)(5 + y) = 700$

Correct Answer: B

Rationale

Choice B is correct. It’s given that it takes the machine **10** minutes to make a large box. It’s also given that ***x*** represents the possible number of large boxes the machine can make each day. Multiplying **10** by ***x*** gives **$10x$** , which represents the amount of time spent making large boxes. It’s given that it takes the machine **5** minutes to make a small box. It’s also given that ***y*** represents the possible number of small boxes the machine can make each day. Multiplying **5** by ***y*** gives **$5y$** , which represents the amount of time spent making small boxes. Combining the amount of time spent making ***x*** large boxes and ***y*** small boxes yields **$10x + 5y$** . It’s given that the machine makes boxes for a total of **700** minutes each day. Therefore **$10x + 5y = 700$** represents the possible number of large boxes, ***x***, and small boxes, ***y***, the machine can make each day.

Choice A is incorrect and may result from associating the time of **10** minutes with small, rather than large, boxes and the time of **5** minutes with large, rather than small, boxes.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.